Table Reduction Document

This document provides a detailed explanation of the process of reducing an entity-relationship model (ERM) to relational tables. It is important to note that some of the steps typically involved in the table reduction process will not be applied due to the way the models were designed.

Elimination of multivalued attributes
NOT APPLICABLE

Elimination of composite attributes
NOT APPLICABLE

All entities from the ERM are converted into tables in the relational model

Each entity from the entity-relationship model is represented as a table in the relational model. Therefore, the minimum number of tables in the resulting relational model will be equal to the number of entities in the ER model.

Transformation of 1:1 cardinality relationships

For each relationship between two entities with 1:1 cardinality, a foreign key is added as an attribute to one of the two entities. The value of this foreign key will be equivalent to the primary key of the related entity.

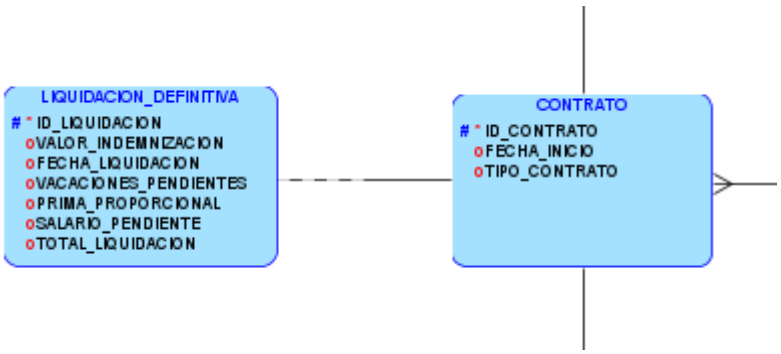


Image 1: Relationship between entities with 1:1 cardinality.

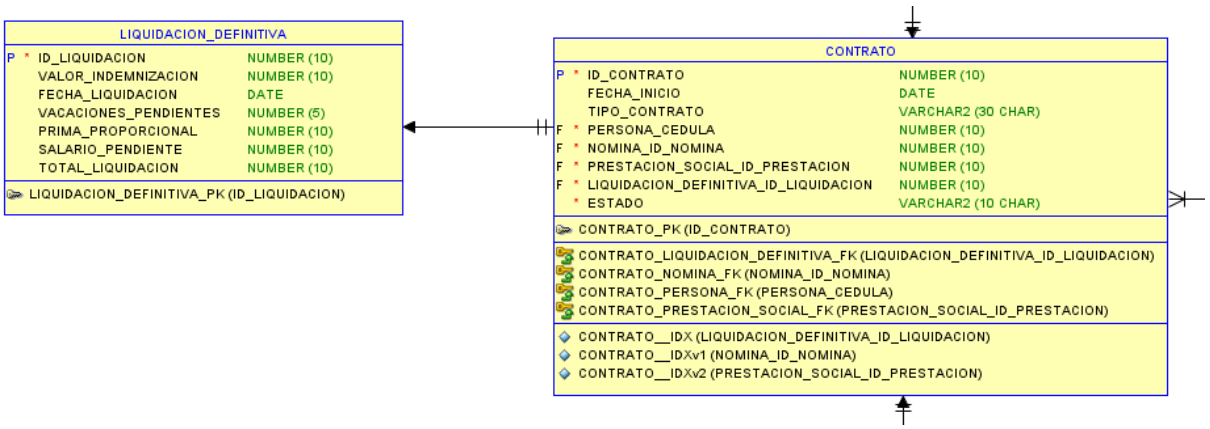


Image 2: Transformation for entities with 1:1 cardinality (foreign key).

Transformation for entities with 1:M cardinality

To transform entities with 1:M cardinality, it is necessary to use a foreign key in a similar manner to the transformation of 1:1 cardinality relationships.

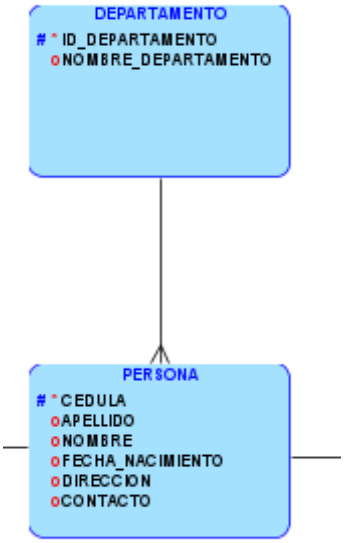


Image 3: Relationship between entities with 1:M cardinality.

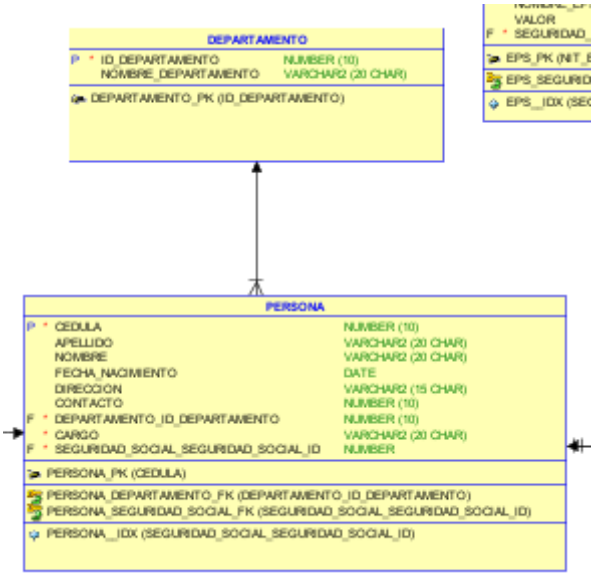


Image 4: Transformation for entities with 1:M cardinality (foreign key).

Transformation for entities with M:N cardinality

To reduce M:N cardinality relationships to tables, it is necessary to "break" the relationship by introducing an intermediate entity that contains the attributes of the original M:N relationship and forms two 1:M relationships with the participating entities. When this intermediate entity is converted into a table, it can be assumed

that its primary key is a composite key that links the tables previously involved in the M:N relationship.

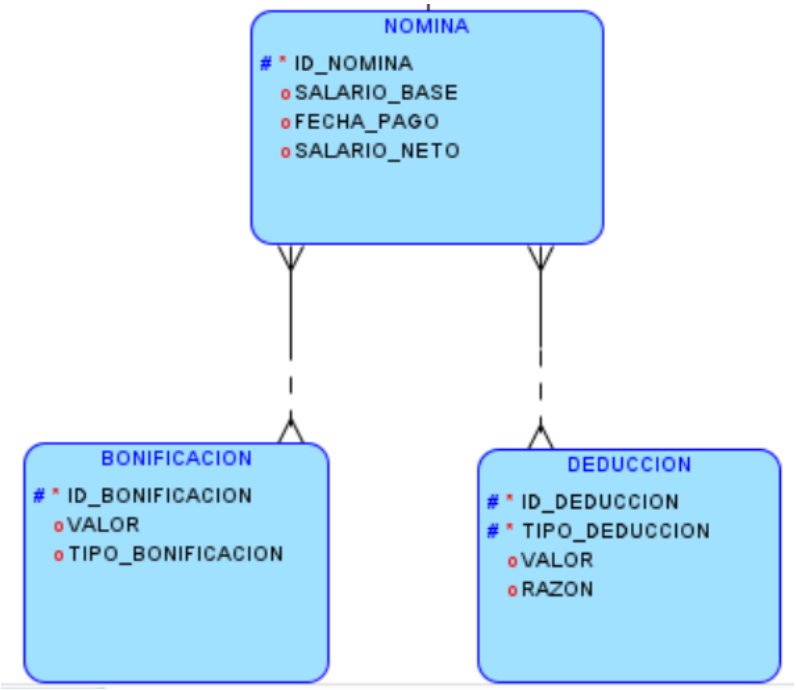


Image 5: Relationship between entities with M:N cardinality.

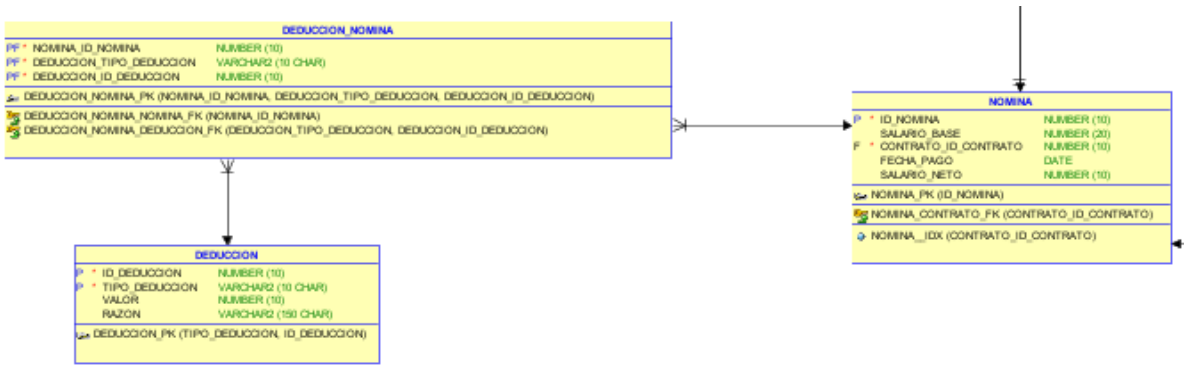


Image 6: Table reduction for relationship with M:N cardinality

Transformation of entities with N-ary relationships
NOT APPLICABLE

Transformation of roles
NOT APPLICABLE

Transformation of generalization/specialization
NOT APPLICABLE

An extended entity-relationship model was not created, as it was not considered entirely necessary for the appropriate design of the database.